



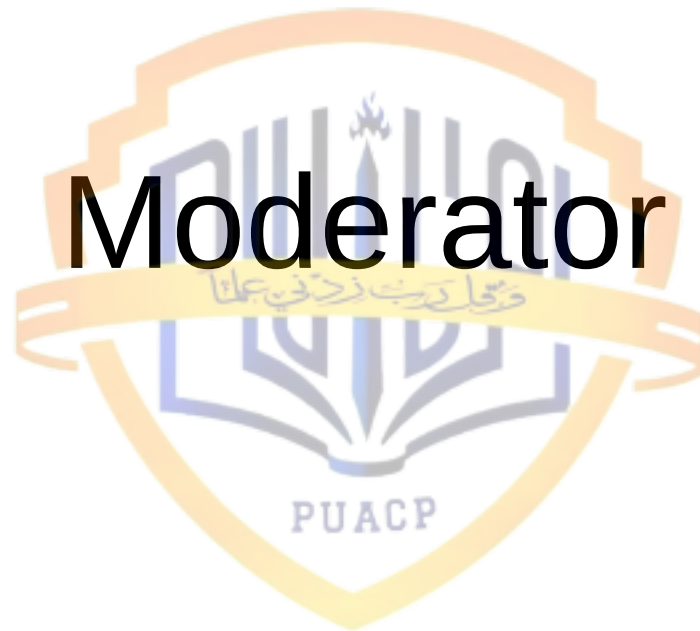
Software Inspections – 6

- Roles

Lecture # 20

Today's Lecture

- Today we are going to discuss different roles played by the participants of the inspection process
- The main emphasis will be on the moderator, however, we'll cover other roles also



Moderator

Moderator - 1

- The moderator is a key role in successful inspections
- He/she ensures that the inspection procedures are followed, that the inspectors fulfill their responsibilities within their defined roles, and that the inspection is performed successfully

Moderator - 2

- A moderator must be trained in the process, principles, and practices of software inspections
- It is not a full-time job, but a part-time assignment, mostly given to senior and experienced programmers, designers, analysts, and writers who have active roles in a project

Moderator - 3

- They should be recognized for the extra time it takes to moderate, whether on their own project or other projects
- The moderator has to play many other tasks including working as a coordinator, facilitator, coach, mediator, manager
- He/she is not a representative of the management

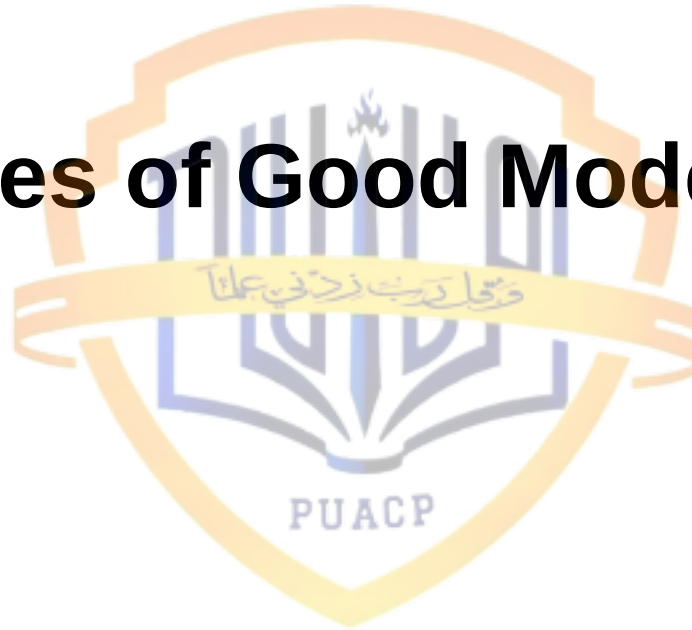
Moderator - 4

- Moderator best serve when they are objective and do not have a vested interest in the work product
- It should be rare to have such a situation
- Moderators best serve when they have technical or domain knowledge of the work product under inspection

Moderator - 5

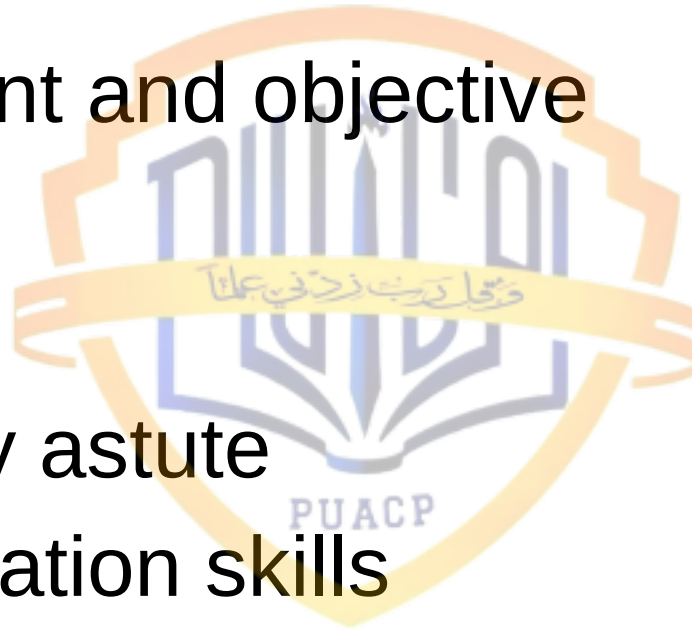
- The moderator should pace the inspection meeting to ensure the participants are not overtaxed, working too long without breaks
- Inspection meetings should not be scheduled for more than two hours
- Moderators must also help in finding defects effectively and efficiently

Qualities of Good Moderators



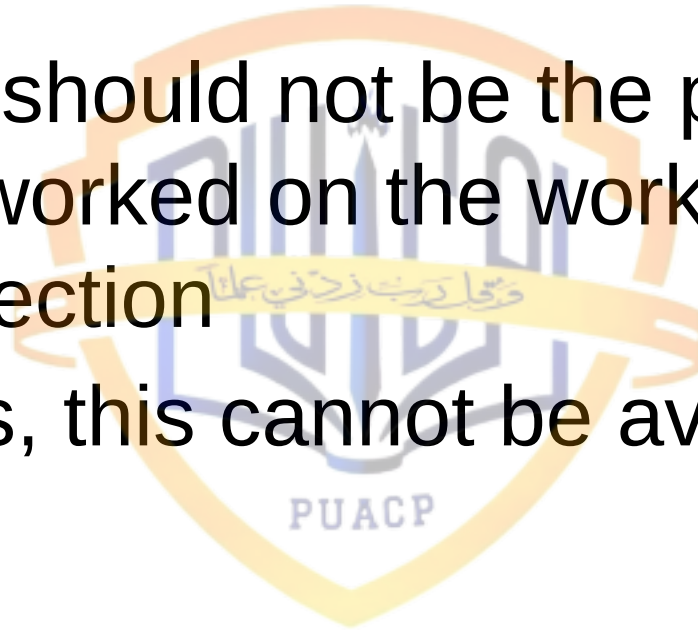
Qualities of Good Moderators

- Independent and objective
- Leader
- Coach
- Technically astute
- Communication skills
- Trained



Independent and Objective

- Moderator should not be the part of the team that worked on the work product under inspection
- Sometimes, this cannot be avoided



Leader

- Moderators serve best when they have management and leadership abilities
- They will manage the inspection once it has been scheduled
- Some organizations have viewed how well a moderator leads as an indication of management ability on future projects

Coach

- Good manager/leaders are also good coaches also

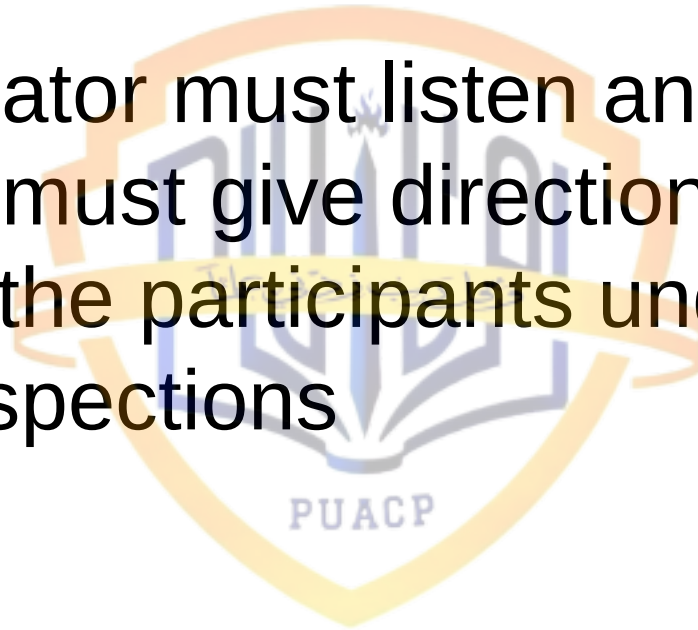


Technically Astute

- The moderator does not have to be an expert in the domain of the work product, but the moderator should be able to understand the technical aspects
- When the moderator is not technically knowledgeable, the team may discount them and they are less able to control the technical discussions

Communication Skills

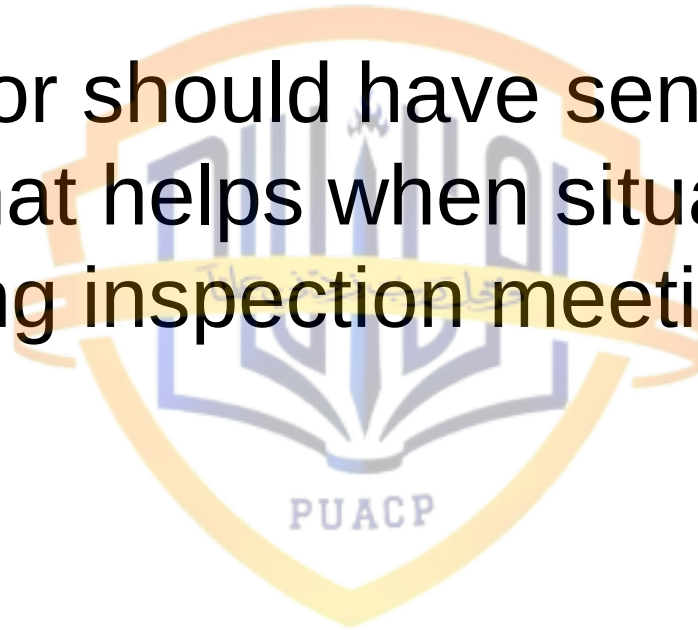
- The moderator must listen and hear; the moderator must give directions and explain so the participants understand the value of inspections



Trained

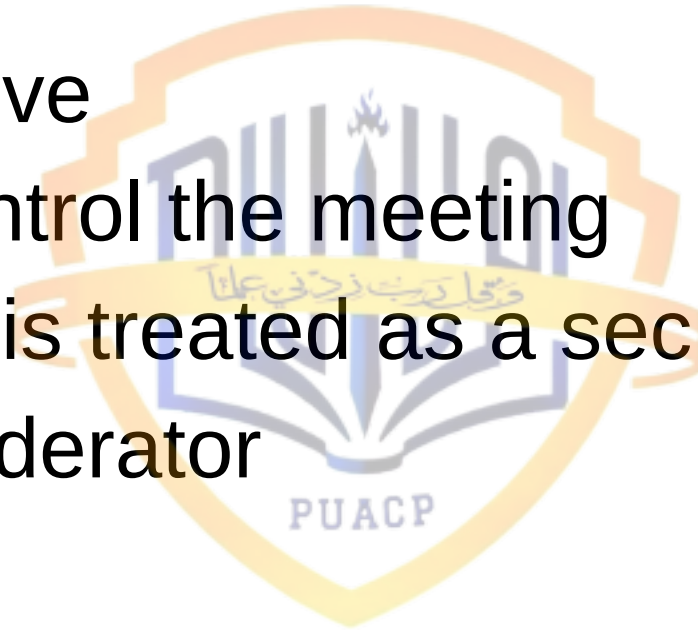
- The moderator must be trained
- Never never have someone serve as a moderator who has not been trained in inspections and the requirements of a moderator

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- A moderator should have sense of humor, because that helps when situation gets tense during inspection meetings



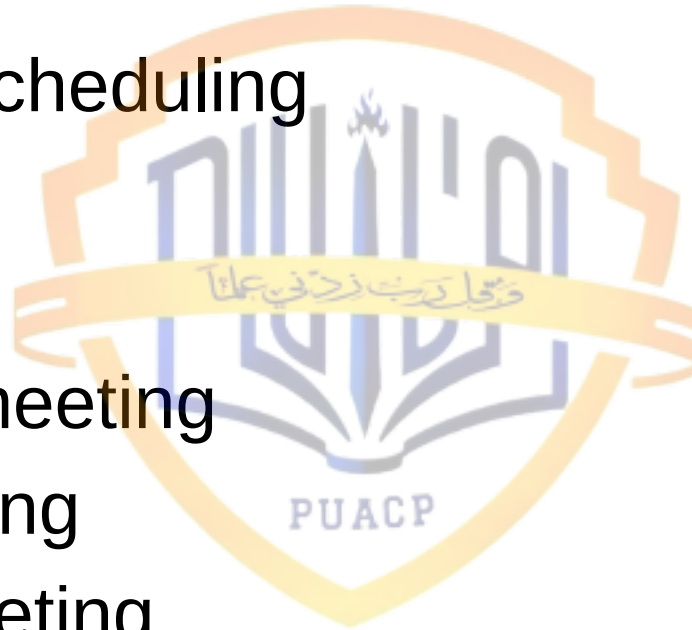
Problems with Moderators

- Is aggressive
- Cannot control the meeting
- Moderator is treated as a secretary
- Biased moderator



Activities to be Performed by the Moderator

- Inspection scheduling
- Overview
- Preparation
- Inspection meeting
- Data recording
- Analysis meeting
- Rework
- Follow-up



Inspection Scheduling - 1

- Determine the need for an overview
- Determine the inspection team
 - Remember that the primary purpose of an inspection is to find the maximum number of defects that may exist in the work product, so pick team members who have the best knowledge and skill to help find defects
- Ensuring availability of materials

Inspection Scheduling - 2

- Assigning roles
- Chunking the materials
 - In situations where multiple meetings are required, split the work product to be inspected into reasonable chunks. This can be done in two ways:
 - By form
 - By function

Inspection Scheduling - 3

- Chunking the material (cont'd)
 - Function chunking, when it is obvious, is easier to do
 - Form chunking is not so obvious, we seek different points of view within documents
 - Standards
 - Code versus other documentation
 - Efficiency
 - User interfaces
 - Maintainability
 - Operating convenience

Inspection Scheduling - 4

- Defining the inspection activities schedule
 - Overview, when required
 - Preparation effort
 - Inspection meeting duration
 - Analysis meeting
 - Logistics

Overview - 1

- Introducing the producer and material for the overview
- Guiding, facilitating, and managing the meeting
- Ensure identified defects that were discovered at the overview are recorded
- Ensure any open issues are recorded

Overview - 2

- Concluding the meeting and asking the participants if the meeting met the objectives
- Overview is conducted by the producer

Preparation

- The moderator as inspector prepares for the inspection meeting just as any other inspection participant



Inspection Meeting - 1

- The moderator has two roles during all inspection meetings
 - Moderator
 - Inspector
- The moderator must always maintain objectivity when serving as an inspector, and there is ample evidence that it can be done

Inspection Meeting - 2

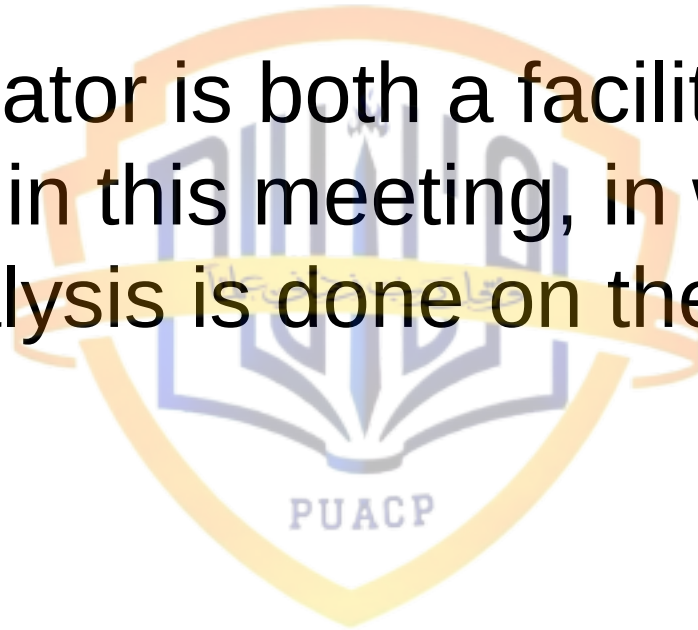
- If moderators hold a mini lessons-learned session at the end of each inspection meeting and ask these questions
 - What worked well
 - What could have been improved
- The inspections process can be improved for future inspection meetings

Data Recording

- The moderator must review the defect report created by the recorder and then complete this report during the follow-up activity for the required contents of the inspection report

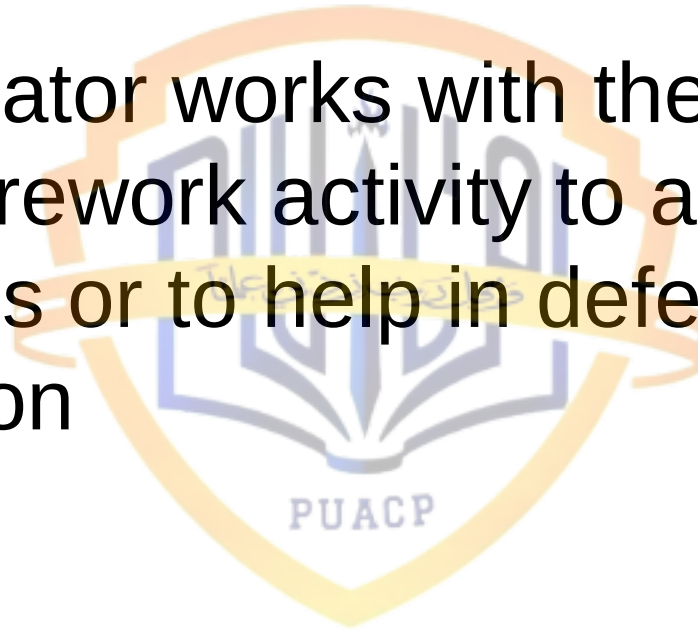
Analysis Meeting

- The moderator is both a facilitator and participant in this meeting, in which a causal analysis is done on the identified defects



Rework

- The moderator works with the producer during the rework activity to address any open issues or to help in defect classification

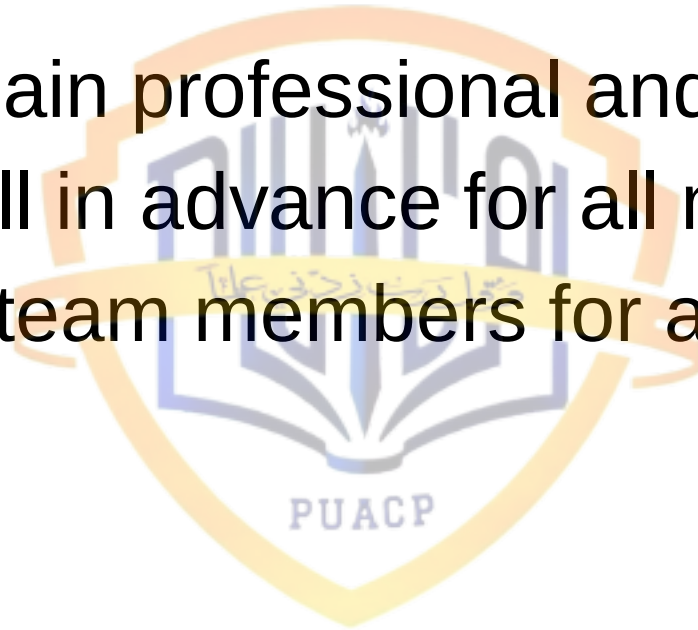


Follow-Up

- Complete the defect report as provided by the recorder to show that the inspection is closed
- Verify all rework (defects and open issues)
- Schedule a re-inspection, if warranted

Code of Conduct for Moderators - 1

- Always remain professional and objective
- Prepare well in advance for all meetings
- Enable the team members for a successful inspection



Code of Conduct for Moderators - 2

- Keep each meeting focused to its specific objectives; e.g.,
 - Learning at the overview
 - Finding and agreeing to defects at the inspection meeting
 - Performing causal analysis at the analysis meeting
- Ensure all data is captured and recorded
- Always maintain confidentiality

Code of Conduct for Moderators - 3

- Use effective meeting practices: e.g.,
 - Properly notify all participants well in advance
 - Restate the purpose of the meeting, especially for first time participants
 - Monitor time and keep the meeting moving forward
 - Allow discussions that help meet the objectives
 - Solicit input at the end of the meeting

Code of Conduct for Moderators - 4

- Be a team player; participate as another inspector
- Remember that the moderator is accountable for the quality of the inspection
- Ensure appropriate behavior by all attendees

Code of Conduct for Moderators - 5

- Enforce and adhere to inspection entry and exit criteria
- Get the consent of the participants to continue the meeting, if it is clear that the inspection meeting will take longer than the scheduled time

Other Roles



Producer

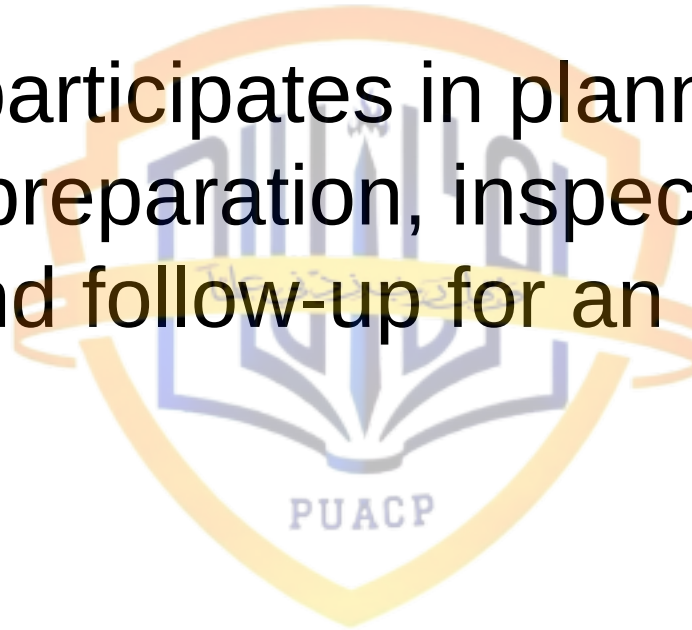


Producer - 1

- The individual who produced or modified the work product to be inspected
- Also known as author
- Producer should be the person who will make changes to the work product as a result of the inspection

Producer - 2

- Producer participates in planning, overview, preparation, inspection meeting, rework, and follow-up for an inspection process



Possible Problems with Producers

- Is defensive
- Does not participate
- Was not the producer
- Responds in a hostile manner to identified defects
- Begins to make repairs at the meeting
- Biased producer
- Unprepared

Producer Types to Watch For

- The Gamesman
- The Controller
- The Intimidator
- The Debater
- The Unbeliever
- The Elitist



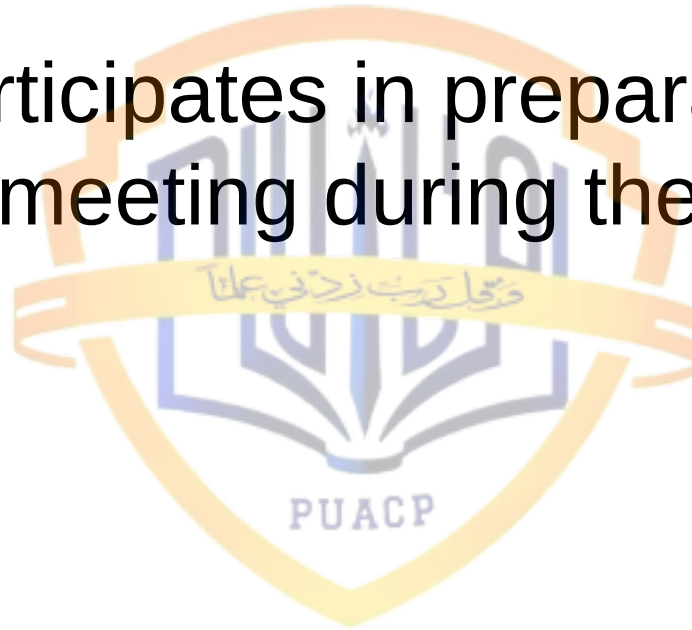


Reader - 1

- The reader is the inspector who will lead the inspection team through the material during the inspection meeting. The purpose of reading is to focus on the inspection material and to ensure an orderly flow for the inspectors

Reader - 2

- Reader participates in preparation and inspection meeting during the inspection process



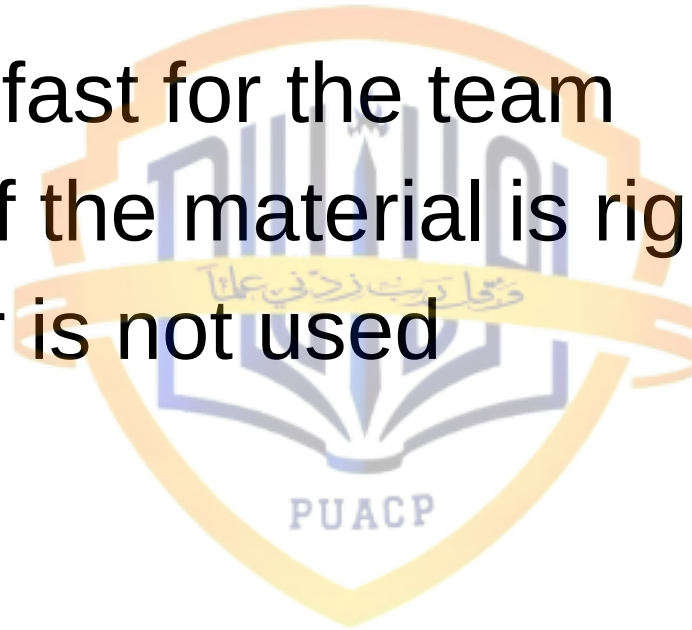
Ways to Read

- Verbatim
- Paraphrase
- Mixed styles
- section-by-section enumeration
- Perspective-based
- Not read



Possible Problems with Readers

- Reads too fast for the team
- Reads as if the material is right
- The reader is not used

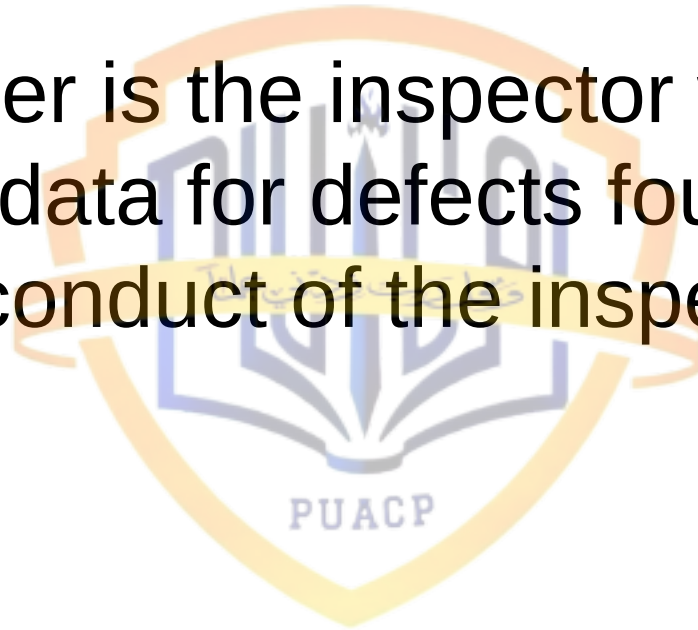




Recorder

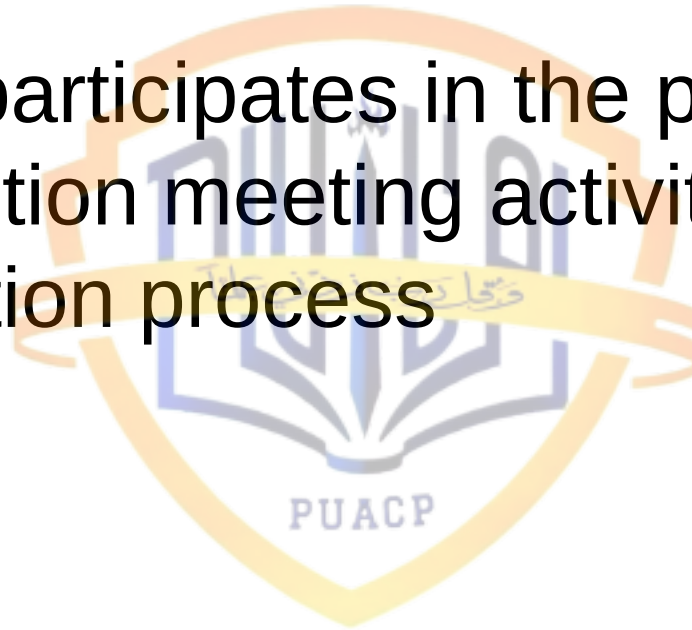
Recorder - 1

- The recorder is the inspector who will record the data for defects found and data about the conduct of the inspection



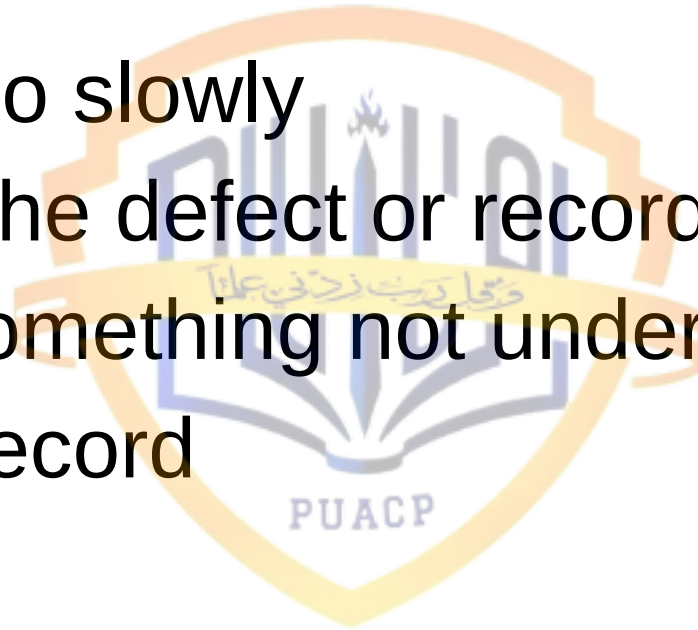
Recorder - 2

- Recorder participates in the preparation and inspection meeting activities during the inspection process



Possible Problems with Recorders

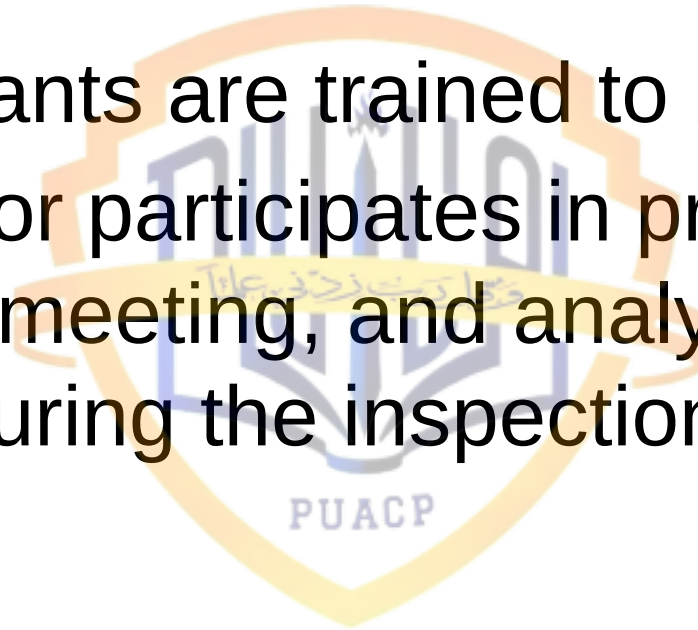
- Records too slowly
- Interprets the defect or records incorrectly
- Records something not understandable
- Does not record





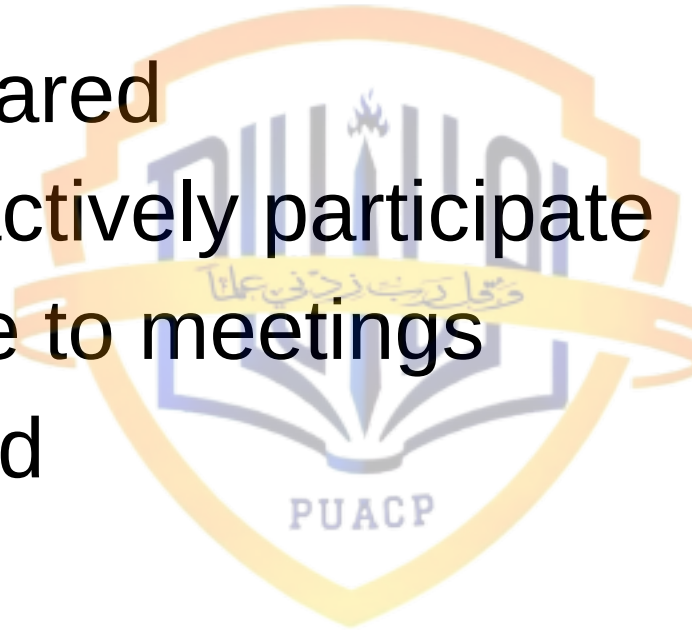
Inspector - 1

- All participants are trained to be inspectors
- An inspector participates in preparation, inspection meeting, and analysis meeting activities during the inspection process



Possible Problems with Inspectors

- Is not prepared
- Does not actively participate
- Comes late to meetings
- Not focused



Criteria for Selecting Inspectors

- Domain knowledge in the work product under inspection
- Experience and expertise
- Language knowledge
- Assignment of inspector with work product
- Time availability
- Trained in inspections
- Team player

Summary

- Today we've focused on different roles used in the inspection process
- Our focus primarily was on the role of the moderator, however, we have covered all other roles as well

References

- Software Inspections by Ronald A. Radice
Tata-McGraw Hill 2003 (Chapter 4-5)

